

READYNEST DATA ANALYTICS INTERNSHIP — WEEK 4 CAPSTONE PROJECT

Predictive Analytics System for Customer Churn Prediction

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| The Problem of Customer Churn

The Core Challenge

Acquiring a new customer is significantly more expensive than keeping an existing one in the telecommunications sector.

When subscribers cancel their plans without warning, it creates immediate revenue losses and impacts long-term business sustainability.

Operational Impact

Organizations often cannot process multi-dimensional account data efficiently to spot high-risk churn signals before users leave.

Without targeted predictions, marketing teams end up spending retention budgets broadly instead of focusing on high-risk accounts.

| Project Objectives



1. Pipeline Assembly

Develop an end-to-end data pipeline in Python to ingest, clean, and pre-process historical customer transaction databases.



2. Machine Learning

Train and compare models (Logistic Regression, Decision Tree, Random Forest) to classify high-risk customer accounts accurately.



3. Dashboard Delivery

Design an interactive 3-page Power BI dashboard to present key metrics, insights, and predictions for management review.

Dataset Overview & Demographics

Parameter	Value / Scope
Data Source	IBM Telco Dataset (Kaggle public archive)
Total Rows	7,043 Customer Profiles
Raw Attributes	21 Column Fields (Demographic, Financial, Services)
Target Variable	Churn (Yes/No) - Binary Category
Class Balance	73.46% Retained / 26.54% Churned

Account Demographics

The dataset contains a rich set of features covering demographic profiles (gender, dependents, senior citizens) and service parameters (internet type, billing options, tech support access).

The uneven split between churn and retained classes matches real-world scenarios and requires specific metric validation checks.

| Data Cleaning & Technical Preprocessing

- ⚠ **Resolving Field Errors:** Fixed formatting errors in the TotalCharges column where 11 entries contained empty string spaces (" ") due to new users having zero-month tenures.
- 📊 **Imputation:** Coerced the empty entries to standard NaN blocks and filled them with the column median value to protect against high-value outliers.
- 🗑 **Cleaning & Encoding:** Checked for and removed duplicate rows. Converted categorical text columns to numeric codes using label mapping and dummy variables.

```
# python data cleaning

df['TotalCharges'] = pd.to_numeric(
    df['TotalCharges'], errors='coerce'
)
df['TotalCharges'].fillna(
    df['TotalCharges'].median(),
    inplace=True
)
df.drop_duplicates(inplace=True)
```


| Exploratory Data Analysis Insights



Contract Configurations

Month-to-month plan users show the highest churn rates, while long-term annual contracts clearly help keep customers loyal.



The Internet Service Paradox

Fiber Optic users churned at a higher rate than DSL subscribers, which correlates closely with premium monthly charges.



Support & Billing

A lack of Online Security or Tech Support options increased churn rates, as did manual payment methods like electronic checks.

Machine Learning Classification Models

ML Algorithm	Accuracy	Features & Attributes
Logistic Regression	80.70%	Sigmoid mapping, excellent calibration, low delay
Random Forest	79.00%	Ensemble pooling, reduces overfitting variance
Decision Tree	74.20%	Partitions rows, prone to training variations

Selected Paradigm

We built and evaluated three distinct models to classify churn. The training was done using standard test-train splits in scikit-learn.

Logistic Regression was selected for production because it provided the highest accuracy, stable probability estimates, and low computational overhead.

Selected Model Performance Details

600 × 400

Confusion Matrix

600 × 400

ROC Curve (AUC = 0.84)

600 × 400

Feature Importances

Our Logistic Regression model achieved **80.70% Accuracy**, an **AUC of 0.84**, and an **Average Precision of 0.63**. Total Charges, tenure, and Monthly Charges emerged as the primary predictive features.

| Interactive Power BI Dashboard Layout

600 × 400

1. Executive Summary

Displays global KPIs: churn rate (26.54%), contract splits, and revenue metrics.

600 × 400

2. Customer Analysis

Tracks demographic trends and monthly charge distributions against user segments.

600 × 400

3. Service Analysis

Maps active utility selections to show how tech support and security options affect loyalty.

Core Business Insights & Recommendations

Identified Attrition Markers

- **Month-to-month Vulnerability:** Account holders on short-term monthly plans show high churn.
- **Price Sensitivity:** High monthly bills, particularly for Fiber Optic lines, increase the risk of churn.
- **Lack of Security/Support:** Skipping Online Security or Tech Support makes accounts less stable.

Recommended Action Items

- ✓ **Incentivize Contract Migrations:** Offer introductory discounts to transition users to annual plans.
- ✓ **Bundle Tech Services:** Package Tech Support and Online Security as basic utility offerings to help stabilize accounts.
- ✓ **Automate Billing Setup:** Drive migrations away from electronic checks toward credit cards or direct debit.

| Conclusion & Future Enhancements

✓ Project Summary

- Cleaned and analyzed the IBM Telco dataset, fixing numeric parsing issues.
- Selected Logistic Regression as the best operational model, achieving 80.70% accuracy.
- Linked predictions to an interactive Power BI dashboard to help guide retention planning.

▶▶ Planned Updates

- Train advanced boosting models (e.g., XGBoost, LightGBM) to capture complex, non-linear relationships.
- Package the model as a live web application using Streamlit.
- Set up real-time stream ingestion to identify customer risk profiles instantly as data is loaded.

Thank You

Questions and Discussion

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